

Question Paper

Exam Date & Time: 03-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

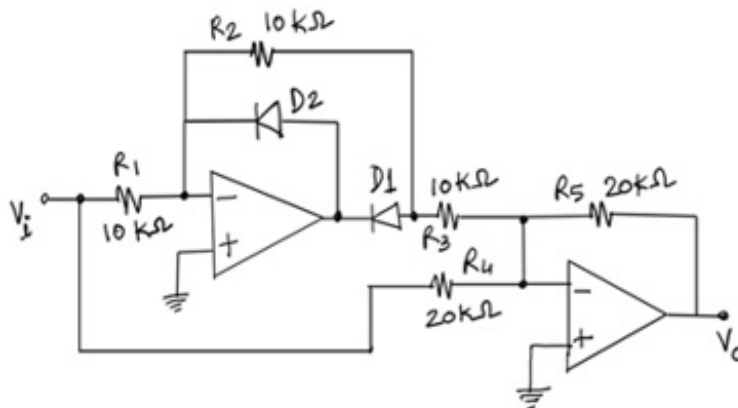
FOURTH SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING) DEGREE EXAMINATIONS -
APRIL / MAY 2025
SUBJECT: ECE 2223 / ECE_2223 - ANALOG INTEGRATED CIRCUITS

Marks: 50

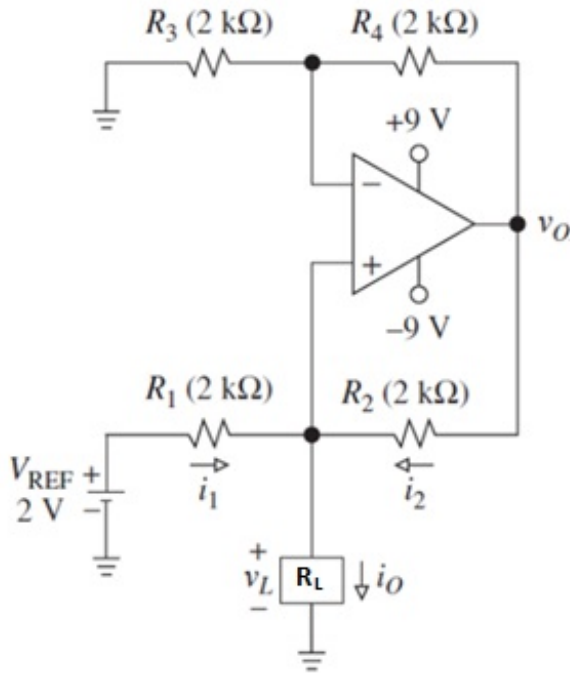
Duration: 180 mins.

Answer all the questions.

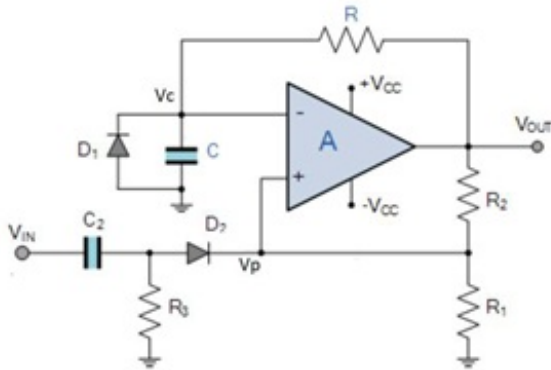
- 1A) Design a circuit using a single op-amp to get the output voltage expression: $V_O = 3V_1 + 4V_2 - 1.5V_3 - 2.5V_4$, where V_1, V_2, V_3 , and V_4 are input voltage sources. Provide the circuit diagram with suitable resistor values. (4)
- 1B) Draw the internal block diagram of op-amp and explain its different stages. (3)
- 1C) i) Draw the circuit used to measure the output resistance of an op-amp and derive the formula for calculating it. (3)
ii) If the output voltage of op-amp is 10 V without a load and drops to 9.52 V when a 1 k Ω load is connected, calculate the output resistance of the op-amp.
- 2A) Design a 4th order low-pass Butterworth filter with a passband gain of 30 dB and cutoff frequency of 2 kHz. Use $C = 0.01 \mu\text{F}$. Determine the roll-off rate in the stopband by computing the gains in decibels at 5 kHz and 50 kHz. (4)
- 2B) For the given circuit, sketch the equivalent circuits corresponding to the positive and negative half-cycles of the input signal. Perform an analysis for each case and draw the resulting output waveform if the input is $5 \sin(314t)$. (3)



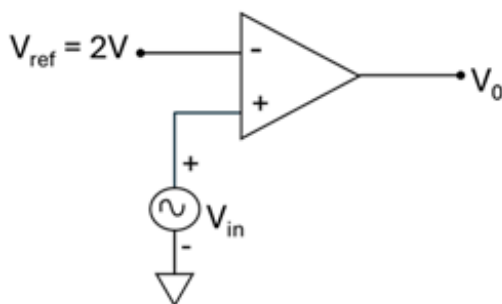
- 2C) Analyze the given circuit and determine the currents i_O , i_1 , and i_2 for load resistances (3)
i) $R_L = 1 \text{ k}\Omega$,
ii) $R_L = 2 \text{ k}\Omega$ and
iii) $R_L = 4 \text{ k}\Omega$.



- 3A) Analyse the following Circuit. Sketch V_P , V_C , and V_{OUT} with respect to trigger signal V_{IN} of pulse with T_1 . Also derive expression for pulse width of the output. (4)



- 3B) An ideal comparator is shown in Figure below. A sinusoidal signal of 8 V peak to peak with the frequency of 1kHz is applied as an input to the comparator. Draw the output waveform and find the duty cycle. (3)



- 3C) Design a first order wide band pass filter having $f_L=400\text{Hz}$ $f_H=2\text{kHz}$ and band pass gain of 4. Draw the frequency response of the filter. (3)
- 4A) With neat circuit diagram and equations explain how 555 timer can be used as FSK generator (4)
- 4B) Derive the 8-bit digital output equivalent to an analog voltage of 5V for a dual slope ADC, which has a maximum integrator output of -6V when the input voltage is 10V (3)
- 4C) Design a 4-bit binary weighted resistor type DAC whose full-scale output voltage is -10V. Logic level 1 = +5 volts and logic level 0 = 0 volt. Determine the output when the input is (3)
- i) 1001 and
ii) 1101.
- What is the resolution of the DAC designed? Select $R = 10\text{k}\Omega$

- 5A) With the neat block diagram and necessary wave form, explain how PLL can be used for AM detection. (4)
- 5B) Explain with the circuit diagram, how LM317 can be used as Adjustable Voltage Regulator? (3)
- 5C) Derive the expression for output voltage of an instrumentation amplifier with suitable diagram. (3)

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Event Name	Subject
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MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - Apr-May 2025 ECE-2223:Analog Integrated Circuits

#	Course Outcome	Description	Marks
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No records found / CO Level Description not defined

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
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No records found / Blooms Level Description not defined